GATE ME 90 Days Challenge

Day 88/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
All Subjects	Mixed PYQ Traps	9 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on the most-repeated GATE traps across the syllabus: successive percentages, counting with a leading zero, limits in degrees, a non-linear ODE that blows up, trace of A^2, kinetic energy of a rolling ring, volumetric strain under uniaxial load, the hollow-shaft k^4 rule, the redundant roller in a cam, static deflection and frequency, Darcy versus Fanning friction factor, local versus average h, Carnot COP in kelvin, I^2R heating and the EOQ holding cost — each solution names the trap so you can avoid it on exam day. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

The price of an article is first raised by 10 % and later the new price is reduced by 10 %. Compared with the original price, the final price is

(A) 1 % lower
(B) the same
(C) 1 % higher
(D) 0.1 % lower

Q2 NAT

The number of three-digit positive integers (100 to 999) in which all three digits are different is _____

Q3 MCQ

If x is measured in degrees, the value of the limit of $(\sin x)/x$ as $x \to 0$ is

(A) 1
(B) $180/\pi$
(C) 0
(D) $\pi/180$

Q4 MCQ

The solution of $dy/dx = y^2$ with $y(0) = 1/2$ is $y = 1/(2 - x)$. Which one of the following statements is correct?

(A) The solution is valid for all real x, and $y(3) = -1$
(B) The solution exists only for $x < 2$; it becomes unbounded as $x \to 2$
(C) The solution exists only for $x > 2$
(D) The equation has no solution through (0, 1/2) because it is non-linear

Q5 NAT

A 3×3 matrix A has the eigenvalues 1, 2 and -3. The trace of A^2 is _____

Q6 NAT

A thin ring (hoop) of mass 2 kg rolls without slipping on a level floor; the speed of its centre is 3 m/s. Its total kinetic energy (in J) is _____

Q7 MCQ

A steel rod 50 mm in diameter and 200 mm long carries an axial tensile load of 100 kN. With E = 200 GPa and Poisson's ratio 0.3, the increase in the volume of the rod is about

(A) 100 mm^3
(B) 160 mm^3
(C) 70 mm^3
(D) 40 mm^3

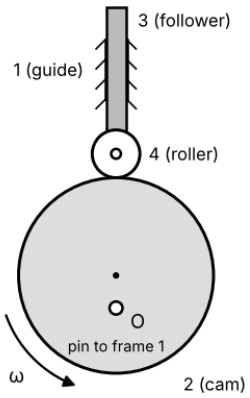
Q8 MCQ

A solid circular shaft is replaced by a hollow shaft of the same material and the same outer diameter D, whose inner diameter is D/2. At the same maximum shear stress, the torque the hollow shaft can carry, as a percentage of that of the solid shaft, is

(A) 75 %
(B) 93.75 %
(C) 50 %
(D) 87.5 %

Q9 MCQ

The cam mechanism shown has a cam (link 2) pinned to the frame (link 1), a roller (link 4) pinned to a follower (link 3) that slides in a frame guide, and rolling (higher-pair) contact between the cam and the roller. Applying Grübler's equation gives F = 2, but the mechanism has only one useful degree of freedom, because



(A) the higher pair removes two degrees of freedom
(B) the free spin of the roller about its own pin is a redundant degree of freedom that does not affect the output motion
(C) the follower guide is a higher pair
(D) Grübler's equation is not valid for mechanisms with sliders

Q10 NAT

A machine on spring mounts has a static deflection of 4 mm. Taking g = 9.81 m/s^2, the natural frequency of its vertical vibration (in Hz) is _____ (round off to 2 decimal places)

Q11 NAT

Oil of kinematic viscosity 0.75 stoke (7.5×10^{-5} m^2/s) flows at a mean velocity of 1.5 m/s through a pipe of 50 mm diameter and 50 m length. Taking g = 9.81 m/s^2, the head lost in friction (in m) is _____ (round off to 2 decimal places)

Q12 NAT

Air flows over a flat plate in laminar flow (local $Nu_x = 0.332\ Re_x^{1/2}\ Pr^{1/3}$). The local heat transfer coefficient at the trailing edge, x = L, is 12 W/m²·K. The average heat transfer coefficient over the whole plate (in W/m²·K) is _____

Q13 MCQ

A reversible heat pump delivers heat to a room at 47 °C while taking heat from the outside air at 7 °C. Its coefficient of performance is

(A) 7.0
(B) 1.18
(C) 8.0
(D) 6.71

Q14 MCQ

In resistance spot welding, the welding current is increased by 20 % while the resistance and the weld time remain the same. The heat generated at the joint increases by about

(A) 20 %
(B) 40 %
(C) 73 %
(D) 44 %

Q15 NAT

The annual demand for a part is 12 500 units. The cost of placing an order is ₹160, the unit price is ₹100 and the annual holding cost is 16 % of the unit price. The economic order quantity (in units) is _____

Answer Key

Q	Type	Answer
Q1	MCQ	(A) 1 % lower
Q2	NAT	648
Q3	MCQ	(D) π/180
Q4	MCQ	(B) The solution exists only for $x < 2$; it becomes unbounded as $x \to 2$
Q5	NAT	14
Q6	NAT	18
Q7	MCQ	(D) 40 mm^3
Q8	MCQ	(B) 93.75 %
Q9	MCQ	(B) the free spin of the roller about its own pin is a redundant degree of freedom that does not affect the output motion
Q10	NAT	7.88 (accept 7.82 to 7.94)
Q11	NAT	7.34 (accept 7.28 to 7.40)
Q12	NAT	24
Q13	MCQ	(C) 8.0
Q14	MCQ	(D) 44 %
Q15	NAT	500

Detailed Solutions

Q1 **Answer: (A) 1 % lower**

Concept: Successive percentage changes multiply; they do not add. Net factor = (1 + a)(1 + b).

Net factor = 1.10 × 0.90 = 0.99.

The final price is **1 % lower (A)** than the original.

Exam tip: Two equal and opposite percentage changes always give a net loss of $(x/10)^2$ %: here $(10/10)^2 = 1$ %. The order of the two changes does not matter.

Q2 **Answer: 648**

Concept: Fill the places one at a time, starting with the most restricted place (fundamental counting principle).

Hundreds digit: 9 choices (1–9, zero not allowed). Tens digit: 9 choices (0–9 except the hundreds digit). Units digit: 8 choices.

Total = 9 × 9 × 8 = **648**.

Exam tip: Starting with 10 × 9 × 8 = 720 forgets that a number cannot begin with 0; 720 − 72 (those starting with 0) = 648 is the same answer by subtraction.

Q3 Answer: (D) π/180

Concept: The standard result $\sin\theta/\theta \to 1$ holds only when θ is in radians.

x degrees $= \pi x/180$ radians, so $\sin x^\circ = \sin(\pi x/180)$.

$\sin(\pi x/180)/x = (\pi/180) \times [\sin(\pi x/180)/(\pi x/180)] \to$ **π/180 (D)** ≈ 0.01745.

Exam tip: The same trap appears in calculus with angles: $d/dx(\sin x^\circ) = (\pi/180)\cos x^\circ$. Always convert to radians before differentiating, integrating or taking limits.

Q4 Answer: (B) The solution exists only for x < 2; it becomes unbounded as x → 2

Concept: For a non-linear equation the solution through a given point may exist only on a limited interval, which must contain the starting point and on which y stays finite.

Separating variables: $-1/y = x + c$; $y(0) = 1/2$ gives $c = -2$, so $y = 1/(2 - x)$.

$y \to \infty$ as $x \to 2^-$, so the solution through $(0, 1/2)$ lives only on $-\infty < x < 2$: **option (B)**. Substituting $x = 3$ into the formula ($y = -1$) gives a point on a different branch, not on this solution.

Exam tip: $dy/dx = y^2$ is always ≥ 0, so a solution cannot drop from $+\infty$ to -1; checking the sign of the slope catches the trap at once.

Q5 Answer: 14

Concept: If λ is an eigenvalue of A, then λ^2 is an eigenvalue of A^2. The trace of any matrix equals the sum of its eigenvalues.

Eigenvalues of A^2: 1, 4 and 9.

$\text{trace}(A^2) = 1 + 4 + 9 =$ **14**.

Exam tip: Squaring the trace, $(1 + 2 - 3)^2 = 0$, is the trap: $\text{trace}(A^2) = \Sigma\lambda^2$, not $(\Sigma\lambda)^2$.

Q6 Answer: 18

Concept: For rolling without slipping, $KE = \tfrac{1}{2}mv^2 + \tfrac{1}{2}I_G\omega^2$ with $\omega = v/r$. For a thin ring $I_G = mr^2$, so the rotational KE equals the translational KE.

Translational KE $= \tfrac{1}{2} \times 2 \times 3^2 = 9$ J; rotational KE $= \tfrac{1}{2} \times (2r^2) \times (3/r)^2 = 9$ J.

Total KE $= 9 + 9 =$ **18 J**.

Exam tip: Writing only $\tfrac{1}{2}mv^2$ (9 J) forgets the spin. The fraction of KE that is rotational: ring 1/2, disc 1/3, solid sphere 2/7.

Q7 Answer: (D) 40 mm³

Concept: Under a uniaxial stress the lateral strains are $-\nu\varepsilon$ each, so the volumetric strain is $\varepsilon_v = \varepsilon(1 - 2\nu)$.

$\sigma = 100\,000/(\pi \times 25^2) = 50.9$ MPa; $\varepsilon = \sigma/E = 2.55 \times 10^{-4}$; $\varepsilon_v = 2.55 \times 10^{-4} \times 0.4 = 1.02 \times 10^{-4}$.

$V = \pi \times 25^2 \times 200 = 3.93 \times 10^5$ mm³, so $\Delta V = \varepsilon_v V =$ **40 mm³ (D)**. Neatly, $\Delta V = PL(1 - 2\nu)/E$, independent of the diameter.

Exam tip: $\Delta V = \varepsilon V$ (100 mm³) ignores the lateral contraction; $(1 + 2\nu)$ instead of $(1 - 2\nu)$ gives 160 mm³.

Q8 **Answer: (B) 93.75 %**

Concept: $T = \tau Z_p$, with $Z_p = \pi D^3(1 - k^4)/16$ for a hollow shaft of diameter ratio $k = d/D$. The material near the centre carries little stress, so removing it costs little strength.

$k = 0.5$: $1 - k^4 = 1 - 0.0625 = 0.9375$.

The hollow shaft carries **93.75 % (B)** of the torque, with only $1 - k^2 = 75$ % of the material (weight).

Exam tip: 75 % (the area ratio) and 50 % (the diameter ratio) are the traps: strength goes with the polar section modulus, which depends on k^4, not on the area.

Q9 **Answer: (B) the free spin of the roller about its own pin is a redundant degree of freedom that does not affect the output motion**

Concept: Grübler: $F = 3(n - 1) - 2j - h$. A redundant degree of freedom is a motion of a link that does not change the input–output relation.

$n = 4$, lower pairs $j = 3$ (cam pin, roller pin, follower slider), $h = 1$: $F = 9 - 6 - 1 = 2$.

The roller can spin on its pin without moving the follower, so the effective DOF $= 2 - 1 = 1$: **option (B)**.

Exam tip: Grübler counts the cam–roller contact as a higher pair (rolling with slipping, $h = 1$). If it is taken as pure rolling without slip, it becomes a lower pair ($j = 4$, $h = 0$) and Grübler gives $F = 1$ directly, with no redundant freedom.

Q10 **Answer: 7.88 (accept 7.82 to 7.94)**

Concept: $\omega_n = \sqrt{(k/m)} = \sqrt{(g/\delta_{st})}$, since $k\delta_{st} = mg$. $f_n = \omega_n/2\pi$.

$\omega_n = \sqrt{(9.81/0.004)} = 49.52$ rad/s.

$f_n = 49.52/(2\pi) =$ **7.88 Hz** (shortcut $f_n \approx 15.76/\sqrt{\delta}$ with δ in mm).

Exam tip: Reporting 49.5 as 'Hz' or putting δ in mm into $\sqrt{(g/\delta)}$ are the two common slips; always check whether rad/s or Hz is asked.

Q11 **Answer: 7.34 (accept 7.28 to 7.40)**

Concept: Darcy–Weisbach $h_f = fLV^2/(2gD)$; for laminar flow the Darcy friction factor is $f = 64/Re$ (the Fanning factor is 16/Re, one-quarter of it).

$Re = VD/\nu = 1.5 \times 0.05/(7.5 \times 10^{-5}) = 1000 \rightarrow$ laminar; $f = 64/1000 = 0.064$.

$h_f = 0.064 \times (50/0.05) \times 1.5^2/(2 \times 9.81) =$ **7.34 m** (Hagen–Poiseuille $32\mu VL/(\rho g D^2)$ gives the same).

Exam tip: Using 16/Re in the Darcy formula gives 1.83 m. If the formula is written with 4f ($h_f = 4fLV^2/2gD$), then f is the Fanning factor.

Q12 **Answer: 24**

Concept: In laminar flow $h_x \propto x^{-1/2}$, so the average over 0 to L is $h_{avg} = (1/L)\int h_x\, dx = 2h_L$ ($Nu_{avg} = 0.664\, Re_L^{1/2}\, Pr^{1/3}$).

$h_{avg} = 2 \times 12$.

$h_{avg} =$ **24 W/m²·K**.

Exam tip: Taking the trailing-edge value as the average underestimates the heat transfer by half. In turbulent flow ($h_x \propto x^{-0.2}$) the factor is 1.25.

Q13 **Answer: (C) 8.0**

Concept: For reversible machines, $COP_{HP} = T_H/(T_H - T_L)$ and $COP_R = T_L/(T_H - T_L)$, with absolute temperatures; $COP_{HP} = COP_R + 1$.

$T_H = 320$ K, $T_L = 280$ K, $T_H - T_L = 40$ K.

$COP_{HP} = 320/40 =$ **8.0 (C)**.

Exam tip: 7.0 is the refrigerator COP; 47/40 = 1.18 uses degrees Celsius. Temperatures in COP and Carnot efficiency must be in kelvin.

Q14 **Answer: (D) 44 %**

Concept: Joule heating: $H = I^2Rt$, so the heat varies as the square of the current.

$H_2/H_1 = 1.2^2 = 1.44$.

The heat rises by **44 % (D)**.

Exam tip: Taking heat $\propto I$ gives 20 %; 2 × 20 % = 40 % is the small-change approximation that misses the 4 % square term. Current is the most powerful control in resistance welding.

Q15 **Answer: 500**

Concept: $EOQ = \sqrt{(2DC_o/C_h)}$, where C_h is the holding cost per unit per year in rupees.

$C_h = 0.16 \times 100 =$ ₹16 per unit per year.

$EOQ = \sqrt{(2 \times 12\,500 \times 160/16)} = \sqrt{250\,000} =$ **500 units**.

Exam tip: Putting $C_h = 0.16$ into the formula gives 5000 units, ten times too large; always convert the holding rate into ₹ per unit per year first.

What next?

Day 88 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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Next, Day 89/90: Final Formula Revision. Comment **GATE90** on our reel to get all links.
Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.